

Optimum design of middle stage tool geometry and addendum surfaces in sheet metal stamping processes using a new isogeometric-based framework

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Abstract

An efficient isogeometric-based framework is presented to integrate optimum design and formability analysis of sheet metal forming processes. To assess the quality of the formed parts, several objective functions such as fracture, wrinkling, thickness variation, and stretching are studied. In this framework, geometric parameters of addendum surfaces and middle tools are considered as design variables, the objective functions are calculated using the recently developed one-step and multi-step inverse isogeometric methods, and the optimum design variables are obtained using the genetic global optimization algorithm. The major advantage of employing the inverse methods is to analyze the formability of the parts with a low computation time. In this research, the effects of altering addendum surfaces and/or middle tools on the quality of the formed parts are simultaneously observed since modeling, formability analysis, and optimization stages of sheet metal forming simulation are integrated using the NURBS functions. To evaluate the performance of the inverse isogeometric models in calculation of the studied objective functions, the results obtained by these models are compared to those of experiment and forward FEM. Comparisons of the results indicate that these models predict the objective functions with acceptable accuracy at a low computation time. For instance, in sheet metal forming analysis of a rectangular box with three different addendum surfaces, the maximum error in prediction of minimum thickness using the one-step inverse model is approximately 4.65% more than forward FEM, while the solution time of forward FEM is around 40 times greater. Finally, the presented optimization procedure is applied to design addendum surfaces in forming of a rectangular box and the middle tools in a two-stage drawing of a square box. The results of these problems confirm the credibility of the present approach in rapid optimum design of addendum surfaces and intermediate tools with acceptable accuracy.

Keywords

Simulation of sheet metal forming processes, inverse isogeometric analysis, addendum design, middle tool design, optimization

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Introduction

Many automobile parts are manufactured by sheet metal forming processes (SMFP) which depend on various parameters. To understand how different parameters can affect the quality of the achieved products, computational approaches, including forward and inverse finite element methods,^{1–8} have been frequently developed in both academic and industrial endeavors. Forward approaches using the incremental theory of plasticity predict more accurate results, compared to inverse models based on the deformation theory of

plasticity. However, a costly trial-and-error process is intrinsically involved in forward methods, in contrast to the inverse models in which reasonably accurate

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results are achieved at a low computational cost. The inputs of inverse models include the geometry of the final part, the initial blank thickness, and material properties of the studied sheet, which are available at the initial design stages.⁵ Because quick estimation of the forming severity is essential in the initial design stages, the inverse methods are frequently used in these stages.

Several research studies^{9–22} were conducted to optimize SMFP, considering different objective functions, design variables, forming solvers, and optimization algorithms. In these works, it was shown that the quality of the formed parts is affected by changing blank-holder/draw-bead forces and geometric parameters of the forming processes. For instance, Debray et al.⁹ demonstrated the significant effects of addendum surfaces on the quality of the formed parts. Therefore, an appropriate design of addendum surfaces can help to reduce fracture and wrinkling problems. In the literature, different objective functions such as fracture, wrinkling, stretching, spring-back, surface aspect, and thickness variation were studied for quality assessment of the formed parts. Also, different design variables including the addendum surface parameters, draw-bead restraining forces, and blank-holder force were selected. Furthermore, computational approaches like forward and inverse finite element methods were utilized to analyze the SMFP, and different optimization methods such as Pareto-based genetic algorithm,^{12,13} genetic algorithm (GA),¹⁰ response surface method (RSM),^{9–12,14,15} and feasible sequential quadratic programming (FSQP)⁹ were implemented to calculate the optimum solutions.

Formability analysis of the formed parts drawn in several stages with considering the middle tools was studied using forward and inverse finite element models in several research studies.^{7,8} However, in multi-step forming processes, the optimum middle tools have not been suggested to achieve the desired quality of the parts so far.

In many research studies done to optimize forming processes, finite element and inverse finite element models were employed to numerically simulate these processes. As FEM-based models separately use modeling and analysis representations, these methods require a long time for preprocessing and mesh generation stages of the simulations. Therefore, implementation of FEM-based models in the iterative optimization procedure demands tremendous computational costs.

To eliminate the gap between the modeling and analysis representations, Hughes et al.²³ proposed isogeometric analysis (IGA). In this methodology, the computer-aided design (CAD) technology is employed. For instance, non-uniform rational B-spline (NURBS) functions are utilized to simultaneously draw and analyze the part in the NURBS-based IGA. This methodology makes it possible to integrate CAD and computer-aided engineering (CAE). Besides, high smoothness of the basis functions and high continuity for high-order basis functions are achieved in this methodology. Therefore, many research studies have

successfully utilized this method in various engineering fields such as shape/topology optimization problems^{24,25} and analysis of shells and plates.^{26,27}

IGA has also been successfully used in simulation of forming processes. For instance, Benson et al.²⁸ studied the stamping process using an isogeometric shell element in a forward incremental approach. In the inverse approach, Zhang et al.²⁹ and Wang et al.³⁰ proposed a one-step inverse isogeometric analysis (IIGA) with and without considering bending effects based on the principle of virtual work. They used the initial solution presented by Zhang et al.⁶ to solve the non-linear governing equations. Furthermore, Shamloofard and Assempour^{31,32} and Isazadeh et al.³³ presented the transfer-based one-step and multi-step IIGA models based on the minimum potential energy. They proved the ability of the transfer-based IIGA models to predict the initial blank and strains in short computation time.

Based on the literature review, the isogeometric methodology has not been applied to optimally design geometric parameters of SMFP so far. Moreover, no attempt has been made to predict the optimum middle tools in multi-stage forming processes. The present study focuses on these important issues. The objective of this research is to develop a new isogeometric-based framework that predicts optimum geometric parameters of the addendum surfaces and the middle stage tools. For this purpose, this procedure takes the advantages of the IIGA models to rapid calculation of the objective and constraint functions using the selected geometric variables. Then, the optimum design variables are computed by employing the GA. In the presented framework, the gap between modeling, analysis, and optimization stages of the simulations is eliminated using the NURBS functions. Therefore, the designer can quickly observe the optimum geometric parameters of the process at the initial design stages.

In what follows, the isogeometric concept and transfer-based IIGA models are briefly reviewed. Next, the capabilities of the IIGA models in calculation of the studied objective functions are experimentally and numerically investigated. Thereafter, a new framework is presented and applied to optimally design SMFP.

Preliminary study

In this section, general concepts of NURBS, IGA, and transfer-based IIGA models are briefly discussed.

NURBS

NURBS and B-spline are the most common parametric functions which are used in CAD. NURBS functions generalized form of B-spline are constructed by imposing weights to B-spline functions. Using a knot vector $E = \{\xi_1, \xi_2, \xi_3, \dots, \xi_{n+p+1}\}$, B-spline basis functions with p degree are recursively generated as follows³⁴:

$$\begin{aligned}
 p = 0 : N_{i,0}(\xi) &= \begin{cases} 1 & \text{if } \xi_i \leq \xi < \xi_{i+1} \\ 0 & \text{otherwise} \end{cases}, \\
 p > 0 : N_{i,p}(\xi) &= \frac{\xi - \xi_i}{\xi_{i+p} - \xi_i} N_{i,p-1}(\xi) \\
 &+ \frac{\xi_{i+p+1} - \xi}{\xi_{i+p+1} - \xi_{i+1}} N_{i+1,p-1}(\xi).
 \end{aligned} \quad (1)$$

NURBS – based IGA

$$\begin{aligned}
 X(\xi, \eta) &= \sum_{i=1}^n \sum_{j=1}^m R_{i,j}^{p,q}(\xi, \eta) X_{i,j}, \\
 V(\xi, \eta) &\approx \hat{V}(\xi, \eta) = \sum_{i=1}^n \sum_{j=1}^m R_{i,j}^{p,q}(\xi, \eta) V_{i,j},
 \end{aligned}$$

Isogeometric analysis (IGA)

The main idea of the NURBS-based IGA is to employ NURBS functions for exact modeling of the parts and approximating field variables. In this methodology, the isoparametric concept is invoked using an exact geometric map, in contrast to FEM that employs an approximate geometric map. In IGA and FEM, a field variable (V) and location of a point (X) are specified as:

Isoparametric FEM

$$\begin{aligned}
 X(\bar{\xi}, \bar{\eta}) &\approx \hat{X}(\bar{\xi}, \bar{\eta}) = \sum_{i=1}^n \sum_{j=1}^m \tilde{N}_{i,j}(\bar{\xi}, \bar{\eta}) X_{i,j}, \\
 V(\bar{\xi}, \bar{\eta}) &\approx \hat{V}(\bar{\xi}, \bar{\eta}) = \sum_{i=1}^n \sum_{j=1}^m \tilde{N}_{i,j}(\bar{\xi}, \bar{\eta}) V_{i,j},
 \end{aligned} \quad (5)$$

In which, $V_{i,j}$, $X_{i,j}$, and $\tilde{N}_{i,j}$ represent the control variables/nodal variables, locations of control points/nodes, and shape functions.

Where, ξ_i and n are i_{th} knot and the number of used basis function/control points, respectively. Considering weights (w_i) and the B-spline basis functions ($N_{i,p}(\xi)$), NURBS basis functions in one-and two-dimensional spaces are defined as³⁴:

$$\begin{aligned}
 R_i^p(\xi) &= \frac{N_{i,p}(\xi)w_i}{\sum_{i=1}^n N_{i,p}(\xi)w_i} \text{ for } i = 1, 2, 3, \dots, n \\
 R_{i,j}^{p,q}(\xi, \eta) &= \frac{N_{i,p}(\xi)M_{j,q}(\eta)w_{i,j}}{\sum_{i=1}^n \sum_{j=1}^m N_{i,p}(\xi)M_{j,q}(\eta)w_{i,j}} \\
 &\text{for } i = 1, 2, 3, \dots, n; j = 1, 2, 3, \dots, m
 \end{aligned} \quad (2)$$

In which, $N_{i,p}(\xi)$ and $M_{j,q}(\eta)$ are single-variable B-spline basis functions related to E and $H = \{\eta_1, \eta_2, \eta_3, \dots, \eta_{m+q+1}\}$ knot vectors.

Using NURBS basis functions ($R_i^p, i = 1, 2, \dots, n$) and control points ($B_i, i = 1, 2, 3, \dots, n$), a NURBS piecewise polynomial curve is obtained as³⁴:

$$C(\xi) = \sum_{i=1}^n R_i^p(\xi) B_i. \quad (3)$$

Similarly, using a control net $\{B_{i,j}\}, i = 1, 2, 3, \dots, n, j = 1, 2, 3, \dots, m$ and NURBS basis functions in two-dimensional space ($R_{i,j}^{p,q}$), a tensor product NURBS surface is constructed as³⁴:

$$S(\xi, \eta) = \sum_{i=1}^n \sum_{j=1}^m R_{i,j}^{p,q}(\xi, \eta) B_{i,j}, \quad (4)$$

Alongside the exact modeling of the parts, the NURBS functions can be successfully used in analysis, due to the unique properties of these functions mentioned by Hughes et al.²³

Inverse isogeometric (IIGA) methodology

Transfer-based one-step and multi-step IIGA models, which are schematically shown in Figure 1, were presented to analyze SMFP^{31,33} with acceptable accuracy at a low computation time. The inputs and outputs of these methods are given in Table 1. These NURBS-based methods deal with the minimization of potential energy, deformation theory of plasticity, and considering isogeometric membrane elements.^{31,33} In these methods, the problem is initially solved by considering elastic conditions, and in the next iterations, plastic

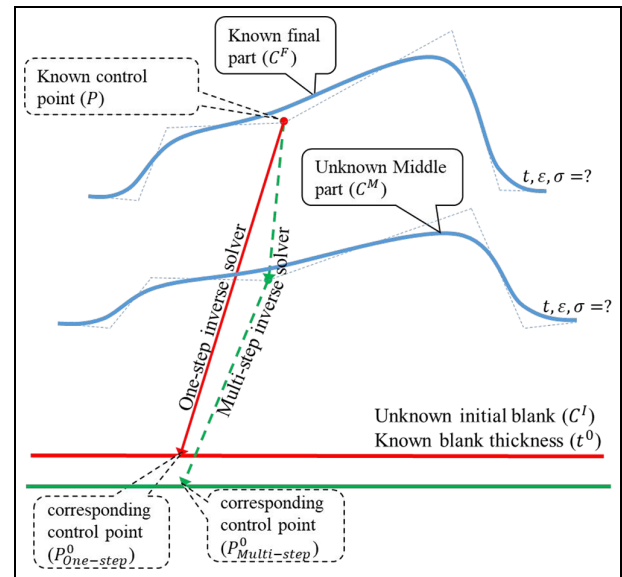


Figure 1. Description of the transfer-based one-step and multi-step IIGA models.

Table 1. Inputs and outputs of the transfer-based IIGA models.

Method	Inputs	Outputs
The one-step IIGA ³¹	The final part, the initial sheet thickness, material properties	Strains/thicknesses of the final part, the shape and size of the blank
The multi-step IIGA ³³	The final part, geometries of the intermediate tools, drawing depth of the middle stage, the initial sheet thickness, material properties	Strains/thicknesses of the final and middle parts, shape and size of the blank, geometry of the middle configuration

properties are added into the computation up to the convergence.

The transfer-based multi-step IIGA model³³ was presented to improve the accuracy of the one-step model in simulations of forming processes that require several forming operations or severe deformation. In these cases, applying the deformation theory of plasticity in several stages and considering information of the middle tools are the main reasons for accuracy enhancement in the multi-step model. This approach is mainly used for analysis of multi-stage forming processes. However, it can be employed to enhance the accuracy of the one-step model in analysis of one-step forming processes.

The aim of this research is far beyond formability analysis of the formed parts with the given geometric parameters. In fact, the objective is to optimally design the geometric parameters of SMFP, by minimizing several objective functions. For this purpose, the one-step and multi-step IIGA models^{31,33} are utilized to calculate the objective/constraint functions in each optimization loop.

The isogeometric-based framework to optimize SMFP

This section aims at presenting an isogeometric-based procedure to design geometric parameters of stamping processes which result in the desired quality of the formed part. This problem can be mathematically stated as the following general form:

$$\begin{cases} \min J_i(\vec{x}), i = 1, \dots, m \\ g_j(\vec{x}) \leq 0, j = 1, \dots, p \\ \vec{x} = \{x_1, x_2, x_3, \dots, x_n\}^T, \\ x_{iL} \leq x_i \leq x_{iU}, i = 1, \dots, n \end{cases} \quad (6)$$

In which, $J_i(\vec{x})$, $g_j(\vec{x})$, $\vec{x}$, x_{iL} , and x_{iU} represent objective functions, constraints, design variable vector, and lower and upper bounds of design variables, respectively. In the next subsections, objective functions, design variables, and optimization algorithm studied in this research are presented.

Objective functions

Among the most important criteria to assess the quality of sheet metal panels, thickness variation, fracture, wrinkling, and stretching are selected in this research.

Thickness variation. In SMFP, thickness variation is a significantly important factor which causes defects in the formed parts. To control this factor, the following function can be minimized⁹:

$$J_T = \frac{1}{ne} \sum_{e=1}^{ne} \left(\frac{t^e - t^0}{t^0} \right)^P \quad (7)$$

Where, t^0 , t , ne , and P are the initial sheet thickness, final sheet thickness, the number of elements, and a positive pair integer. Several research studies^{9,15} stated that it is severe to include the above objective function in the optimization processes. Therefore, in this research, thinning and thickening constraints are applied to the optimization process, according to the following inequality:

$$-\Delta_{min} \leq \frac{t^e - t^0}{t^0} \leq \Delta_{max} \quad (8)$$

In which, Δ_{min} and Δ_{max} are permissible values of minimum and maximum thickness strains.

Fracture and wrinkling. The risk of fracture and wrinkling during the forming process can be evaluated by situating the principal strains on the forming limit diagram (FLD).³⁵ In this diagram, shown in Figure 2, the fracture or wrinkling problems may occur if the principal strains are located above the forming limit curve or below the wrinkling curve. To minimize these defects, the following functions are defined:

$$\begin{cases} J_F = \left(\sum_{e=1}^{ne} (\epsilon_1^e - \epsilon_F(\epsilon_2^e))^P \right)^{1/P} & \text{if } \epsilon_1^e > \epsilon_F(\epsilon_2^e) \\ J_W = \left(\sum_{e=1}^{ne} (\epsilon_W(\epsilon_2^e) - \epsilon_1^e)^P \right)^{1/P} & \text{if } \epsilon_1^e < \epsilon_W(\epsilon_2^e) \\ J_F = J_W = 0 & \text{Otherwise} \end{cases} \quad (9)$$

In which, J_F , J_W , ϵ_1 , ϵ_2 , ϵ_F , and ϵ_W represent fracture and wrinkling functions, principal strains, and equations of fracture and wrinkling curves according to Figure 2, respectively.

Stretching. Inadequate stretch of the formed part is an important concern which decreases the dent resistance

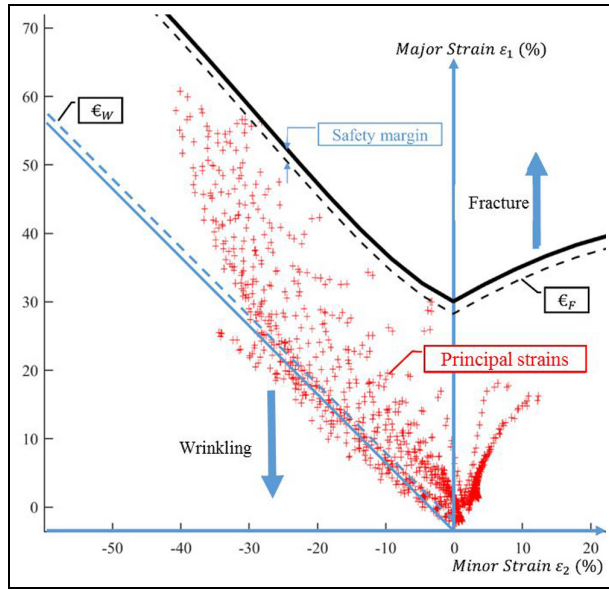


Figure 2. Illustration of principal strains on the FLD.

of the part.³⁶ To reduce this problem, effective plastic strain in each element should be greater than a specific value. Mathematically, the following function is defined to minimize insufficient stretching of the formed part^{12,13,15}:

$$J_S = \begin{cases} \sum_{e=1}^{ne} (\bar{\epsilon}_{min} - \bar{\epsilon}^e)^2 & \bar{\epsilon}^e < \bar{\epsilon}_{min} \\ 0 & \bar{\epsilon}^e > \bar{\epsilon}_{min} \end{cases} \quad (10)$$

Where, J_S , $\bar{\epsilon}$, and $\bar{\epsilon}_{min}$ are the objective function of insufficient stretching, effective strain of each element, and a minimum acceptable effective strain which is considered between 0.03 and 0.05.¹²

Design variables

As stated earlier, geometric parameters of addendum surfaces and the middle tools are considered as design variables in this study.

Addendum surface parameters. Figure 3(a) shows a part manufactured by a forming process. In this part, addendum surfaces, which will be cut after the forming process, include the protection wall, die profile radius, and blank-holder surfaces. Figure 3(b) displays an addendum profile curve. Considering the parameters D , R_d , β , R_h , and H at nodes located on the trim line of the desired part, profile curves are generated as the following equations:

$$\text{If } \beta < \alpha \quad \begin{cases} x_Q = x_O + R_h \cos(\beta) \\ y_Q = y_O + R_h \sin(\beta) \end{cases}, \quad \begin{cases} x_O = x_P - R_h \cos(\alpha) \\ y_O = y_P - R_h \sin(\alpha) \end{cases}$$

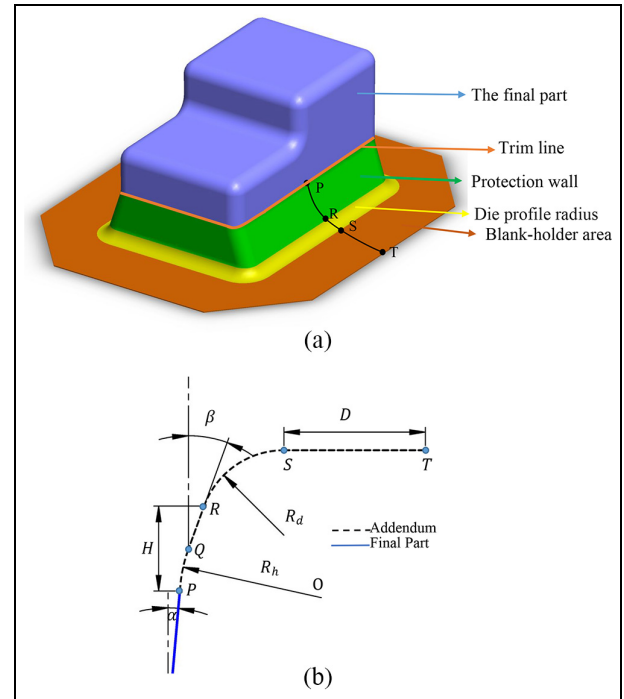


Figure 3. (a) A final part and its addendum surfaces and (b) the addendum profile curve.

If $\beta > \alpha$

$$\begin{cases} x_Q = x_O - R_h \cos(\beta) \\ y_Q = y_O - R_h \sin(\beta) \end{cases}, \quad \begin{cases} x_O = x_P + R_h \cos(\alpha) \\ y_O = y_P + R_h \sin(\alpha) \end{cases}$$

$$\begin{aligned} x_R &= x_Q + (H - (y_Q - y_P)) \tan(\beta) \\ y_R &= y_Q + (H - (y_Q - y_P)) \\ x_S &= x_R + R_d \cos(\beta) \\ y_S &= y_R + R_d(1 - \sin(\beta)) \\ x_T &= x_S + D \\ y_T &= y_S \end{aligned}$$

(11)

The above equations are solved in each optimization loop for all points located on the trim line, and therefore the entire part is updated for the next step.

The middle tools' parameters. The quality of the formed parts is affected by varying the middle tools of SMFP. In this research, the shape and size of the tools, their angles, and fillet radii in the middle forming stage are considered as design variables. Also, to limit the variety of design variables, only several standard shapes of tools are taken into consideration.

Figure 4 shows the geometric variables of the middle tools in the two-stage forming of a square box. Considering the design variables D^m , R_p^m , θ^m , and R_d^m , the middle part is constructed similar to equation (11).

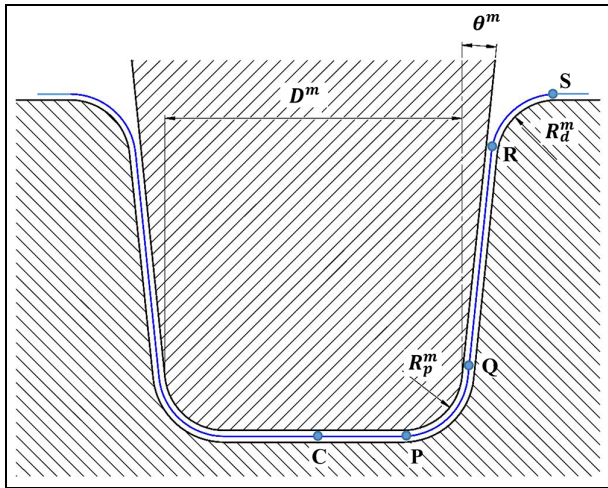


Figure 4. The geometric variables of the middle tools.

The relationship between physical and control points. In the previous sections, locations of physical points which are connected to the addendum surfaces and middle parts are computed in terms of design variables (D, R_d, β, R_h, H) and $(D^m, R_p^m, \theta^m, R_d^m)$. After specifying physical points, locations of control points should be calculated since the IIGA models require control points to analyze the process. To do so, the following equation is used³³:

$$\begin{Bmatrix} x(\xi, \eta) \\ y(\xi, \eta) \\ z(\xi, \eta) \end{Bmatrix} = \begin{bmatrix} R_1 & \cdots & R_{ncp} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & R_1 & \cdots & R_{ncp} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 0 & R_1 & \cdots & R_{ncp} \end{bmatrix} \times \{X_1, \dots, X_{ncp}, Y_1, \dots, Y_{ncp}, Z_1, \dots, Z_{ncp}\}^T \quad (12)$$

Where, (x, y, z) , (X, Y, Z) , and ncp represent the coordinates of physical points, control points, and the number of control points in each isogeometric element, respectively.

Solving procedure

Figure 5 shows the sequences of the presented isogeometric-based framework to optimize SMFP. In this framework, locations of physical and control points in each optimization loop are calculated, according to the selected design variables. In the subsequent steps, the IIGA solvers are employed for formability analysis of the part, and therefore the objective functions and constraints are obtained. Eventually, design variables are updated, and the process is repeated until the GA³⁷ converges.

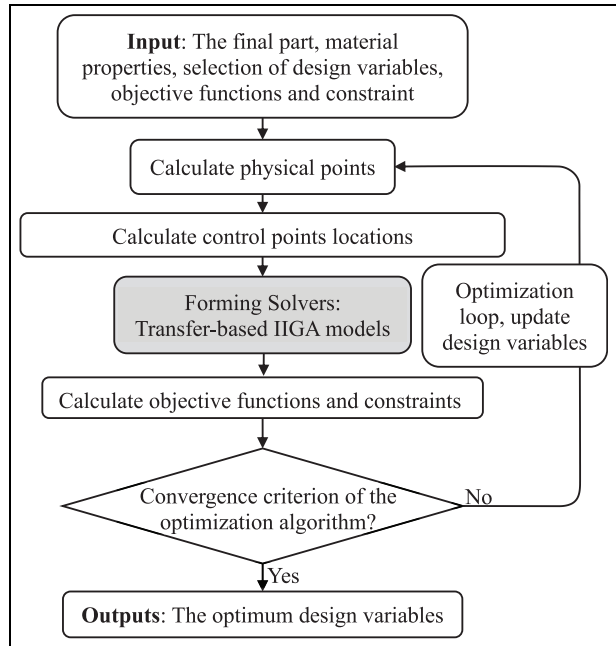


Figure 5. Flow diagram of the present framework.

Results and discussions

To evaluate the credibility of the presented framework, this approach is initially applied to design addendum parameters of a rectangular box formed in one operation. Next, geometric parameters of the middle tools in a two-stage drawing of a square box are designed so that the desired quality of the final part can be obtained.

Problem I: Addendum design of a rectangular box

The purpose of this problem is to design addendum parameters of a rectangular box shown in Figure 6(a) with considering the following objective and constraint functions:

$$\begin{cases} \min(0.5 \times \bar{J}_W(\vec{x}) + 0.5 \times \bar{J}_S(\vec{x})), \\ \text{Subject to :} \\ J_F(\vec{x}) = 0; \\ -0.25 < \frac{t(\vec{x}) - t^0(\vec{x})}{t^0(\vec{x})} < 0.25 \\ \vec{x} = \{H, \beta, R_h, R_d, D_1, D_2, D_3\} \end{cases}$$

With

$$\begin{aligned} 0 &\leq H \leq 6, 5 \leq \beta \leq 15, 5 \leq R_h \leq 10, 5 \leq R_d \leq 10, \\ 0 &\leq D_1 \leq 23, 0 \leq D_2 \leq 40, 0 \leq D_3 \leq 22 \end{aligned}$$

In which, $(\bar{J}_W)$ and $(\bar{J}_S)$ indicate the normalized wrinkling and stretching functions. Also, the studied addendum parameters $\{H, \beta, R_h, R_d, D_1, D_2, D_3\}$ are shown in Figure 6(b).

Validation and selection of the forming solver. Before using the one-step and multi-step IIGA models^{31,33} in the optimization process, the performance of these

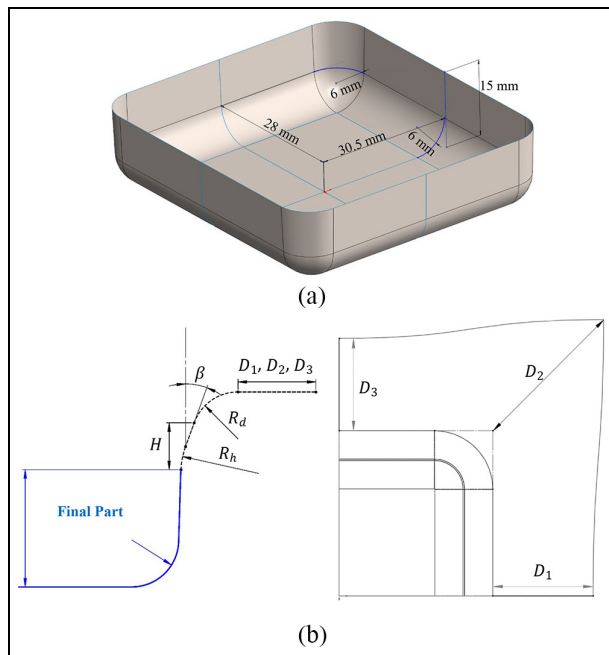


Figure 6. (a) The final part in problem I and (b) addendum parameters in problem I.

methods is evaluated in formability analysis of the final part with considering different addendum surfaces. To do so, the results of IIGA models are compared to those of forward FEM and experiment.

To obtain experimental results, three square blanks with dimensions of $128\text{ mm} \times 128\text{ mm}$ and thicknesses of 1 mm were formed up to 16, 20, and 26 mm of drawing depths. Using a tensile test, the mechanical properties and hardening behavior of the utilized sheets were acquired according to ASTM E8 and Hollman's power law, respectively. Accordingly, the sheet material properties and forming conditions are given in Table 2. A 30-ton hydraulic press at the velocity of 5 mm/s was also employed to form the aforementioned parts. The parts acquired from the experiments, shown in Figure 7, were measured by the VMM machine (YF-2010D + Touch Probe) and then the measured information was imported into the IIGA models to analyze the processes. Also, forward finite element analysis was simulated by using the standard implicit solver of the ABAQUS software and 40×40 four-node shell elements.

Table 3 compares the initial blanks resulted from the one-step IIGA, multi-step IIGA, and experiments. As revealed in this table, the blanks are predicted with acceptable accuracy in the IIGA models. Also, thicknesses along paths A and B, shown in Figure 7, for all approaches are presented in Figure 8. In the experimental approach, thicknesses of the final parts along these paths were measured by a micrometer after cutting the parts. These paths are selected because the minimum and maximum thicknesses of the entire part are located along paths A and B. Compared to the experimental results, the maximum errors in predictions of minimum

Table 2. Material properties and forming conditions.

Parameter	Value
Strength coefficient	538.6 MPa
Strain hardening exponent	0.26
r_0, r_{45}, r_{90}	1.322, 1.132, 1.733
Lankford coefficient (r)	1.330
Friction coefficient (μ)	0.1
Blank holder force	10 kN
Punch dimension	61×56
Die dimension	63.6×58.6
Initial sheet thickness	1 mm
Punch/die radius	6 mm

and maximum thicknesses are approximately 8.39%, 8.42% for one-step IIGA, 7.99%, 7.23% for multi-step IIGA, and 3.74%, 2.54% for forward FEM. Comparisons of results disclose that forward FEM predicts more accurate results than IIGA models. Also, IIGA models predict thicknesses of the final parts with acceptable accuracy. The solution time for solving these tests using forward FEM and the multi-step IIGA model is around 40 and six times greater than the time needed in the one-step model. Thus, the one-step IIGA solver is selected to optimally design addendum surfaces in this problem.

Solving the optimization problem. After verification and selection of the forming solver, the most effective addendum parameters should be specified. To study how the considered addendum parameters can affect the objective function, analysis of variance was conducted. To do so, the Taguchi test³⁸ plan with 27 orthogonal arrays (L27) was employed in which seven parameters at three levels (1, 2, and 3) were considered. In the Taguchi test plan, levels 1, 2, and 3 of a parameter indicate minimum, average, and maximum values of this parameter in the studied domain. In all 27 tests, the whole parts consisting of the final part and addendum surfaces were generated and then imported into the one-step IIGA model. In the Taguchi test plan, the importance of each parameter is evaluated by the signal-to-noise ratio (S/N). In this problem, this ratio was calculated for minimizing the objective function. The calculated S/N ratios for the seven addendum parameters are presented in Table 4. The range is defined as the difference between maximum and minimum values of S/N ratio for each parameter. In Table 4, the higher rank demonstrates which parameters are more influential on the objective function. So, in this problem the most effective addendum parameters are D_1, D_3, D_2 , and H . Also, to decrease the number of independent parameters, D_1 and D_3 were related to each other by considering the following constraint:

$$D_3 = r_p D_1$$

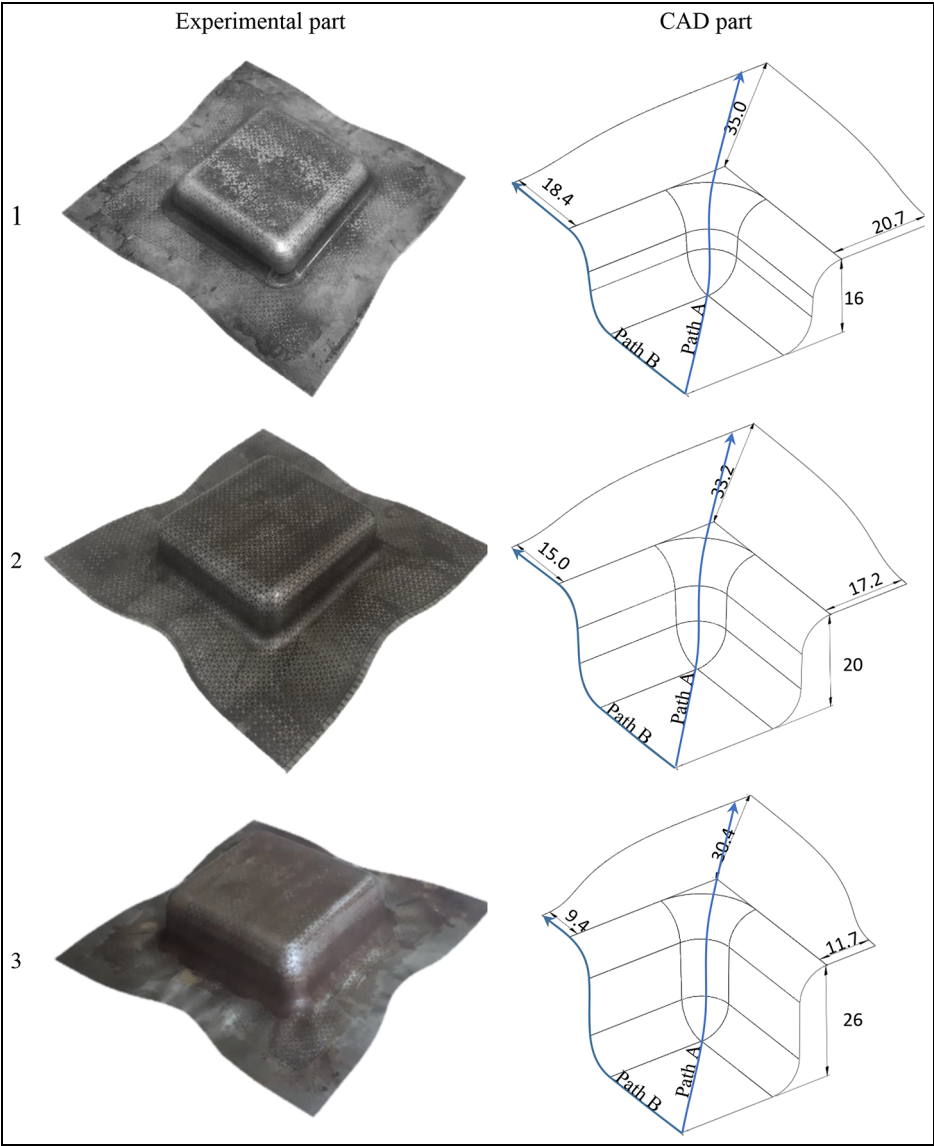


Figure 7. The parts obtained by experiments (dimensions in mm).

Table 3. Comparison of the initial blanks in problem I.

Test number	One-step IIGA model	Multi-step IIGA model	Experiment
	Max error in prediction of the initial blank (%)	Max error in prediction of the initial blank (%)	The initial blank
Test #1	1.334	1.305	128mm×128mm
Test #2	1.394	1.35	128mm×128mm
Test #3	1.789	1.679	128mm×128mm

Where, r_p is the punch dimension ratio and its value is equal to $\frac{56}{61}$, according to Table 2.

Therefore, D_1 , D_2 , and H are finally considered as design variables of the optimization algorithm and the problem is rewritten as:

$$\begin{cases} \min(0.5 \times \bar{J}_W(\vec{x}) + 0.5 \times \bar{J}_S(\vec{x})), \\ \text{Subject to:} \\ J_F(\vec{x}) = 0; \\ -0.25 < \frac{t(\vec{x}) - t^0(\vec{x})}{t^0(\vec{x})} < 0.25 \\ \vec{x} = \{H, D_1, D_2\} \end{cases}$$

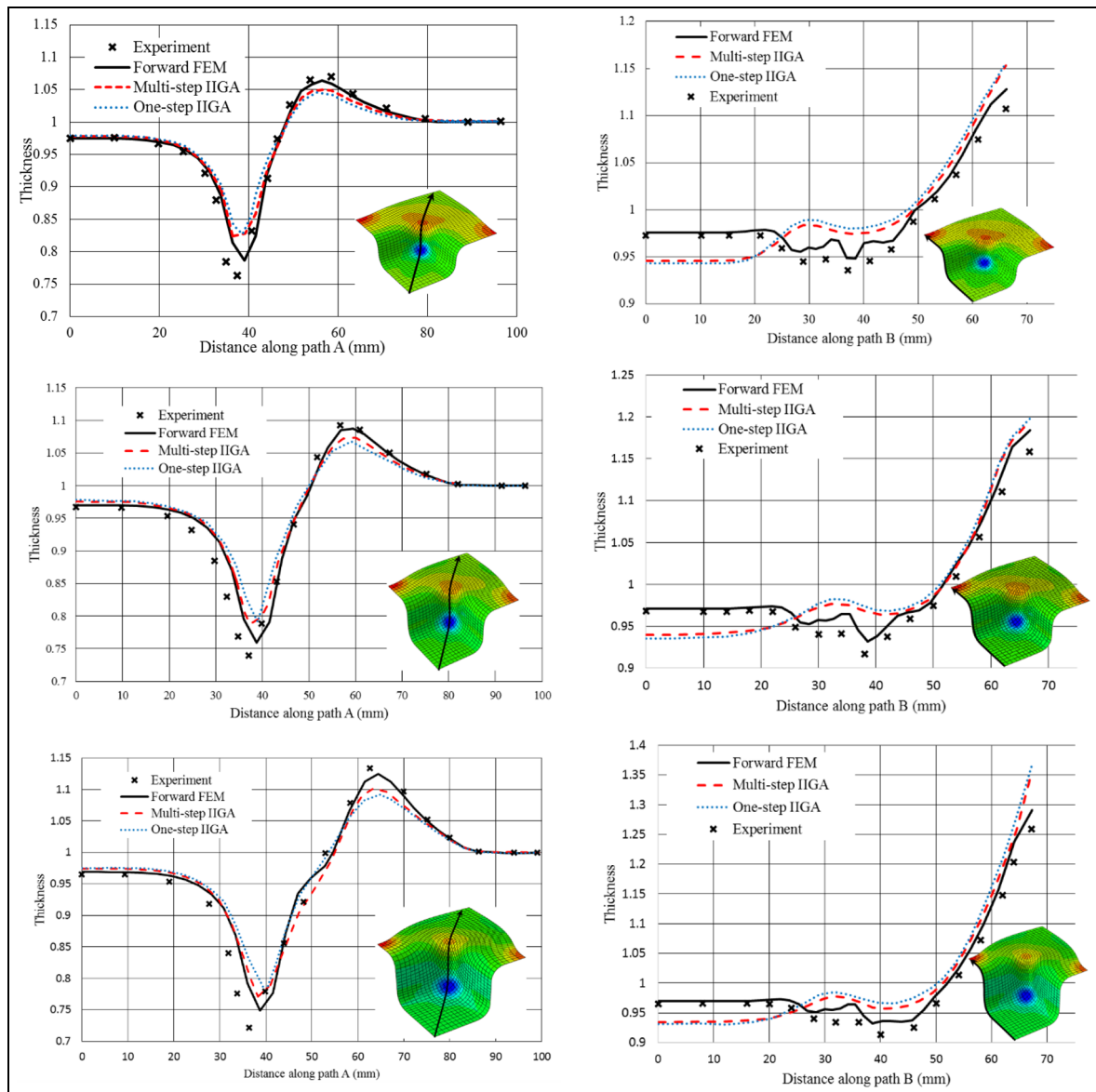


Figure 8. Comparison of thickness distribution along paths A and B in the studied tests in problem I.

Table 4. Analysis of variance.

	H	β	R_h	R_d	D_1	D_2	D_3
1	5.179	4.773	4.294	4.919	7.364	5.815	6.913
2	4.756	4.685	4.647	4.597	4.466	3.711	4.881
3	4.228	4.584	5.011	4.525	2.212	4.621	2.247
Range	0.951	0.19	0.717	0.394	5.151	2.104	4.666
Rank	4	7	5	6	1	3	2

With

$$0 \leq H \leq 6, 0 \leq D_1 \leq 23, 0 \leq D_2 \leq 40$$

The above optimization problem was solved using the presented framework. The calculated optimum answer is shown in Figure 9.

In this research, using the fast IIGA solvers, which remove the gap between modeling and analysis representations, makes it possible to optimally design geometric parameters of SMFP with acceptable accuracy at low computation time. Table 5 presents the required CPU time to calculate the optimum answer with an

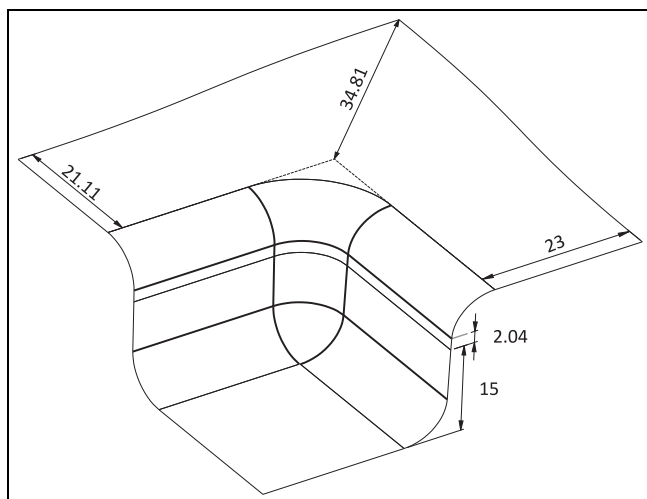


Figure 9. The final part with optimum addendum surfaces (dimensions in mm).

Intel® Core™ i7-3632QM CPU, using two different initial populations in the GA.³⁷ Obviously, the solution time for solving this optimization problem exponentially increases if FEM-based and/or incremental-based methods are used as the forming solver of the GA.

Figure 10 displays the principal strains calculated by the one-step IIGA model on the FLD for the calculated optimum addendum surface and two other addendum surfaces, which lead to unsuitable wrinkling and stretching conditions. This figure shows that the major improvements in the quality of the final part are obtained by considering the optimum addendum parameters, compared to the others. In Figure 10(a), it is demonstrated that the wrinkling problem is significantly reduced by selecting appropriate addendum surfaces. Moreover, as displayed in Figure 10(b), the equivalent strains of 610 elements are smaller than 0.03 in case of considering inappropriate addendum parameters. This number is noticeably decreased in the optimum case (71 elements), and therefore the insufficient stretching problem is meaningfully reduced. Thus, the appropriate design of addendum surfaces improves the quality of the part significantly.

Figure 11 compares the required blanks to form the optimum part and the final parts with minimum and maximum addendum parameters. As revealed in this figure, the material consumption is decreased in the optimum case, compared to the upper bound of addendum parameters. Therefore, optimum design of

addendum surfaces not only can enhance the quality of the part, but also can save the material-consuming.

Problem 2: Design of the middle tools in two-stage forming of a square box

In this problem, optimum geometric parameters of the middle tools in two-stage drawing of a square box depicted in Figure 12 are designed with considering the following objective function:

$$\begin{cases} \min(\bar{J}_W(\vec{x}) + \bar{J}_F(\vec{x})), \\ \text{Subject to :} \\ \vec{x} = \{S, A^m, \theta^m, R_p^m\} \end{cases}$$

With

$$S = \{\text{Circle} - \text{Square} - \text{Rectangle}\}, 5548\text{mm}^2 \leq A^m \leq 9503\text{mm}^2, 5^\circ < \theta^m < 40^\circ, 6\text{mm} \leq R_p^m \leq 12\text{mm}$$

In which, S , A^m , θ^m , and R_p^m represent the shape and size of the middle punch, punch angle, and punch profile radius shown in Figure 13. Also, $(\bar{J}_F)$ indicates the normalized fracture function. To limit the variety of design variables, only standard punch shapes, including circle, square, and rectangle are taken into considerations. The material properties of the used sheet are given in Table 2.

Validation and selection of the forming solver. The results of forward FEM, IIGA methods, and suggestion of the sheet metal forming handbook³⁹ indicate that it is impossible to form this square part in one operation. Therefore, the multi-step IIGA model³³ should be employed as the forming solver of the optimization procedure.

Prior to using this solver, the credibility of this method was evaluated in sheet metal forming analysis of the final part drawn in two operations with different middle tools. For this purpose, two-stage forming analysis of the square part with two different middle tools was carried out using the multi-step IIGA and forward FEM. The middle tools utilized in these tests are shown in Figure 14. Considering the middle tools of the tests, the multi-step IIGA solver was initially used for formability analysis of the final part. The blanks obtained by this method are shown in Figure 15. These blanks as well as the considered middle tools of each test were imported into the forward FEM and incremental analysis of these tests was conducted. The final parts resulted

Table 5. The required time to achieve the optimum answer in problem I.

Initial population	Number of crossover in each iteration	Number of mutation in each iteration	Number of iteration up to convergence	The required time up to convergence
50	40	20	25	≈15hr
100	80	40	20	≈24hr

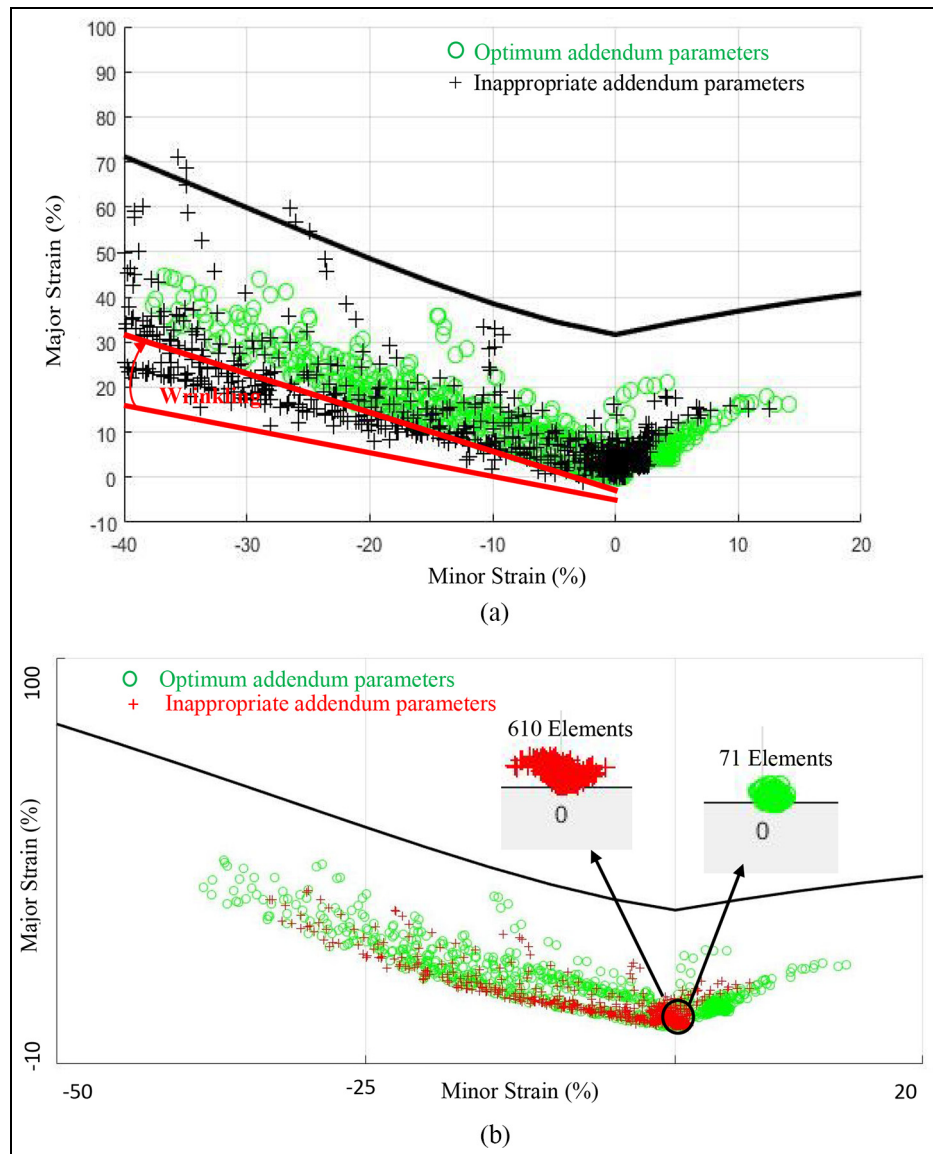


Figure 10. Illustration of the principal strains on the FLD for the final part with different addendum surfaces: (a) improvement of the wrinkling problem and (b) improvement of the insufficient stretching problem.

from forward FEM and multi-step IIGA are compared in Figure 16. As depicted in this figure, the maximum deviation in prediction of the final parts obtained by forward FEM and multi-step IIGA is approximately 1.56 and 4.4 mm. This amount of error is acceptable, due to the dimensions of the final part. Thicknesses along the diagonal path of the final parts in these tests are presented in Figure 17. The similar trend and acceptable consistency are observed in both approaches. Although forward finite element analysis of these severe forming processes leads to more accurate results, the optimum design of middle tools using this approach requires tremendous computation time and powerful computers. In contrast to forward FEM, the multi-step IIGA rapidly predicts the objective function with acceptable accuracy.³³ Therefore, the multi-step IIGA model is selected to optimally design the middle tools in this problem.

Solving the optimization problem. After verification of the multi-step IIGA solver, the optimization problem was solved using the algorithm described in Figure 5. The optimum middle part resulted from this process is displayed in Figure 18. This figure reveals that the best choice for the middle punch shape is the circular one. This is exactly the middle punch shape suggested in the sheet metal forming handbook³⁹ for deep drawing of a square box.

In Figure 19, the principal strains obtained by the multi-step IIGA model in case of using different middle tools are displayed on FLD. The considered middle tools include the optimum answer and the circular/rectangular middle tools which result in the unsuitable quality of the final part. This figure shows that the optimum design of middle tools decreases the wrinkling and fracture problems. To illustrate the effects of the middle tools on the wrinkling reduction of the final

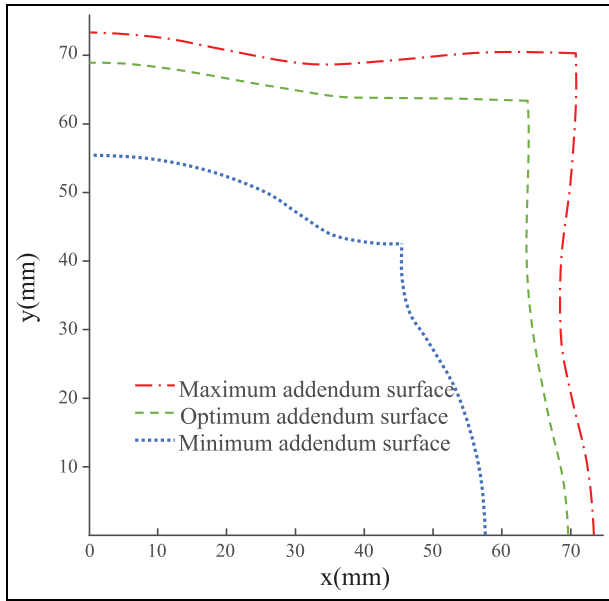


Figure 11. Comparison of the initial blanks required to form the final part with considering different addendum surfaces.

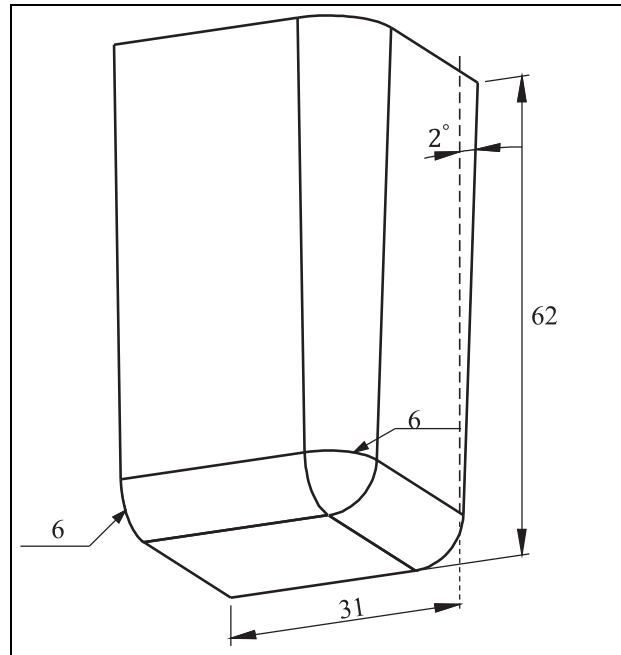


Figure 12. A quarter of the final part in problem 2 (dimensions in mm).

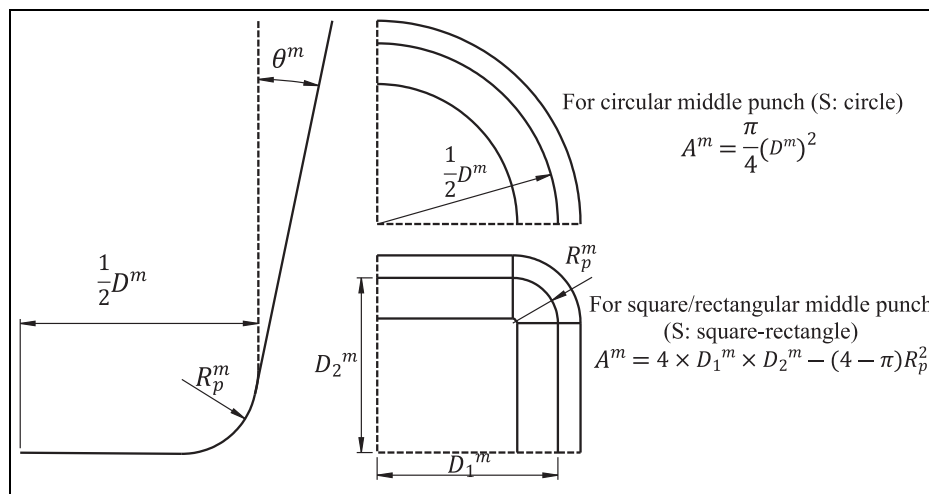


Figure 13. Geometric parameters of the middle punch.

part, the parts obtained by forward FEM are shown and compared in Figure 20 for two different cases, (a) employing the optimum middle punch (Figure 18) calculated by the developed procedure and (b) using the circular middle punch ($A^m = 7864 \text{ mm}^2$, $\theta^m = 6.5^\circ$, $R_p^m = 9.5 \text{ mm}$, according to Figure 13) which leads to improper wrinkling condition. Comparison of the parts in Figures 20 shows the important effects of middle tools on the quality enhancement of the final part.

Conclusion

In this study, the IGA-based optimization framework was developed to integrate modeling, analysis, and optimization steps of the simulations using the NURBS

functions. This framework takes advantages of the fast IIGA methods to calculate the objective/constraint functions of the genetic global optimization method. Using the IGA-based approach makes it possible to simultaneously observe the effects of changing geometric parameters of the forming processes on the quality of the final parts. The presented optimization procedure was successfully applied to optimally design addendum surfaces of the rectangular box and middle tools of the square box formed in two operations. The results of the optimization process disclose that the wrinkling, fracture, insufficient stretching defects of the formed parts are meaningfully decreased by the appropriate design of addendum surfaces, and middle tools. In addition, rapid optimum design of the SMFP with

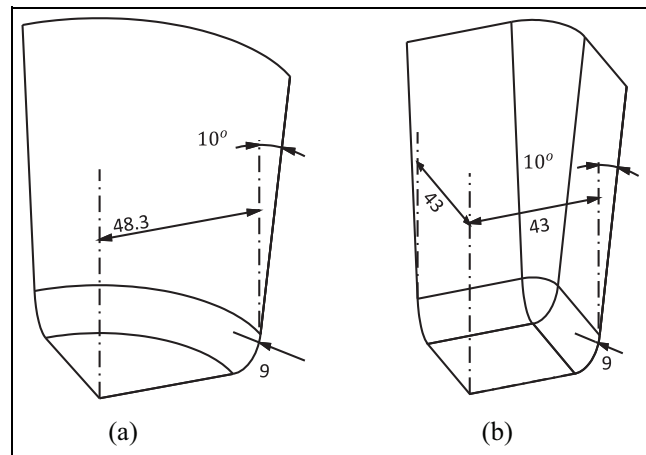


Figure 14. A quarter of the middle tools utilized in the verification tests of problem 2: (a) the circular middle punch of test 1 and (b) the square middle punch of test 2 (dimensions in mm).

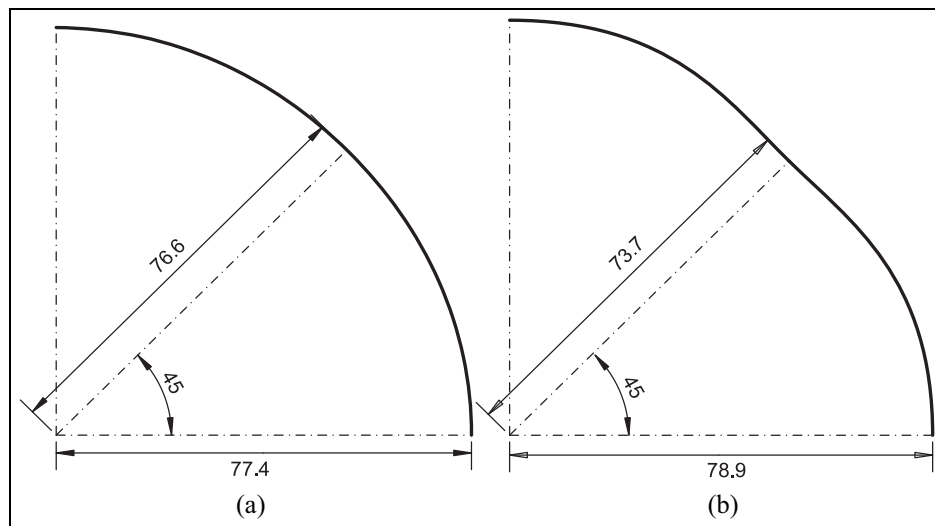


Figure 15. A quarter of the initial blanks obtained by multi-step IIGA in the verification tests of problem 2: (a) test 1 and (b) test 2.

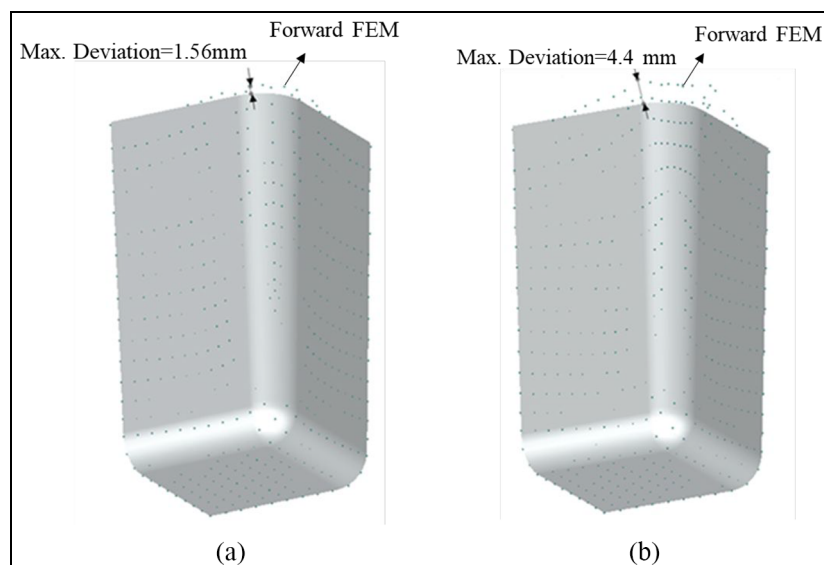


Figure 16. Comparison of the final parts obtained by forward FEM and multi-step IIGA (a quarter of the actual parts is demonstrated): (a) test 1 and (b) test 2.

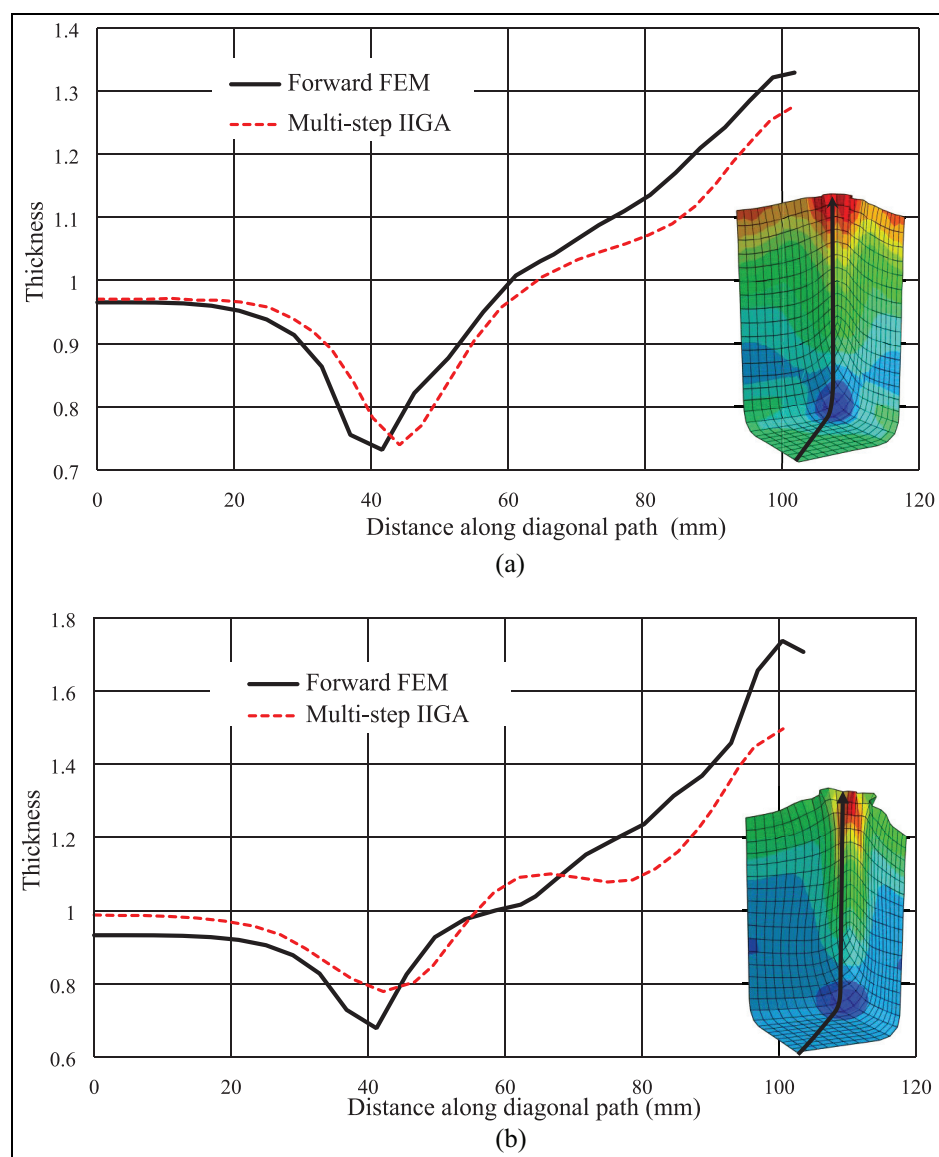


Figure 17. Comparison of thicknesses along the diagonal path of the final parts in the verification tests of problem 2: (a) test 1 and (b) test 2.

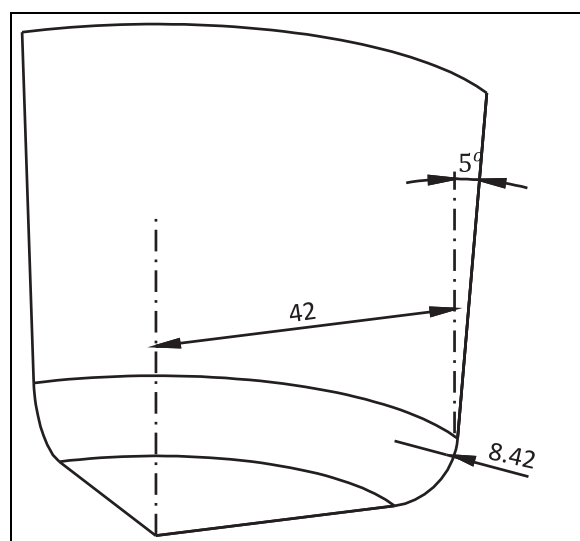


Figure 18. The optimum middle punch resulted from the presented framework (dimensions in mm, a quarter of the actual part is demonstrated).

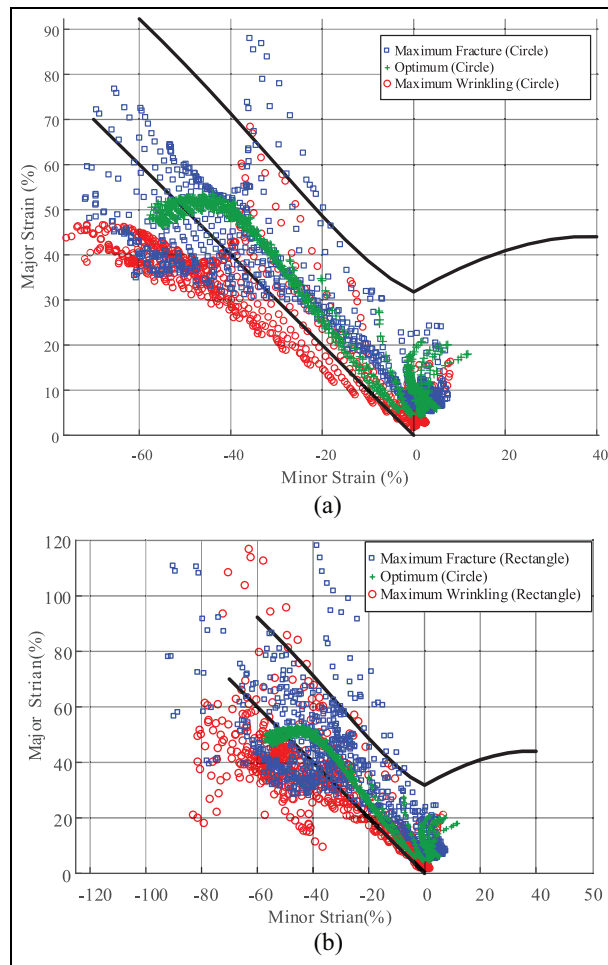


Figure 19. Illustration of the forming severity of the final part with considering different middle tools: (a) optimum middle tools and circular middle tools which result in inappropriate quality and (b) optimum middle tools and rectangular middle tools which result in inappropriate quality.

acceptable accuracy is achieved using the developed IGA-based procedure.

Declaration of conflicting interests

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References

1. Majlessi SA and Lee D. Further development of sheet metal forming analysis method. *J Eng Ind* 1987; 109(4): 330–337.
2. Guo YQ, Batoz JL, Naceur H, et al. Recent developments on the analysis and optimum design of sheet metal forming parts using a simplified inverse approach. *Comput Struct* 2000; 78(1–3): 133–148.
3. Yang DY, Kim TJ and Yoon SJ. Recent trends in numerical simulation of three-dimensional sheet forming processes. *Proc IMechE, Part B: J Engineering Manufacture* 2003; 217(11): 1553–1570.
4. Fazli A and Arezoo B. Optimum blank shape design using shape error and deformation path length of

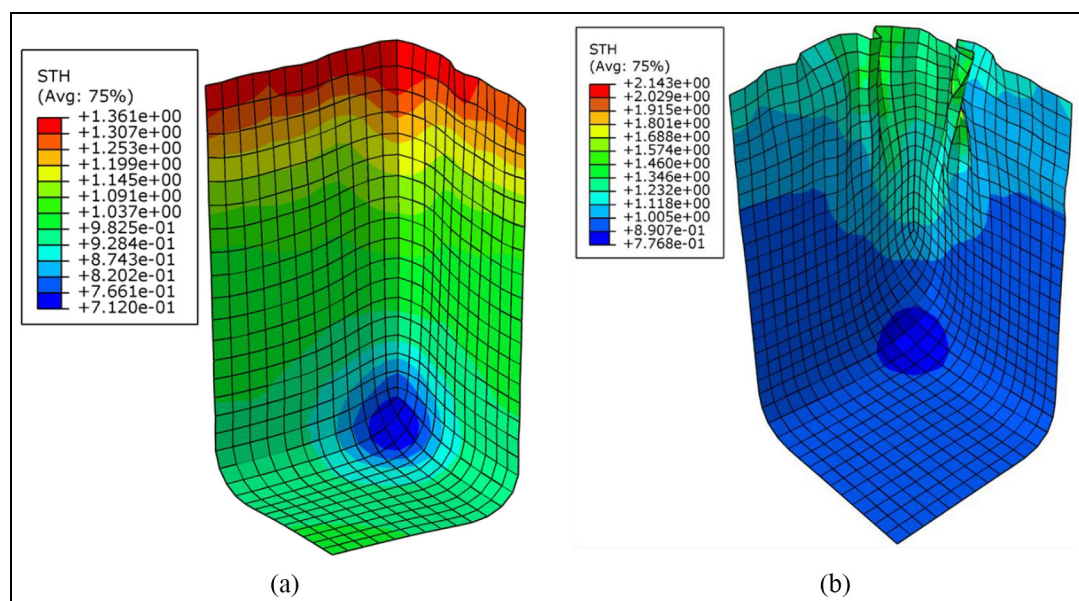


Figure 20. The final parts resulted from forward FEM with considering two different middle tools: (a) the calculated optimum middle tools and (b) the circular middle tools which lead to improper wrinkling condition.

- boundary nodes. *Proc IMechE, Part B: J Engineering Manufacture* 2012; 226(4): 684–693.
5. Bostan Shirin M and Assempour A. Some improvements on the unfolding inverse finite element method for simulation of deep drawing process. *Int J Adv Manuf Technol* 2014; 72(1–4): 447–456.
6. Zhang X, Hu S, Lang Z, et al. Energy-based initial guess estimation method for one-step simulation. *Int J Comput Methods Eng Sci Mech* 2007; 8(6): 411–417.
7. Bostan Shirin M and Assempour A. A new mapping technique to consider intermediate stages in multistep deep drawing. *Proc IMechE, Part C: J Mechanical Engineering Science* 2014; 228(9): 1557–1569.
8. Huang Y, Chen YP and Du RX. A new approach to solve key issues in multi-step inverse finite-element method in sheet metal stamping. *Int J Mech* 2006; 48(6): 591–600.
9. Debray K, Li YM and Guo YQ. Parametric design and optimization of addendum surfaces for sheet metal forming process. *Int J Mater Form* 2013; 6(3): 315–325.
10. Hu W, Enying L, Li GY, et al. Optimization of sheet metal forming processes by the use of space mapping based metamodeling method. *Int J Adv Manuf Technol* 2008; 39(7–8): 642–655.
11. Chebbah MS, Naceur H and Hecini M. Rapid coupling optimization method for a tube hydroforming process. *Proc IMechE, Part B: J Engineering Manufacture* 2010; 224(2): 245–256.
12. Wei L and Yuying Y. Multi-objective optimization of sheet metal forming process using Pareto-based genetic algorithm. *J Mater Process Technol* 2008; 208(1–3): 499–506.
13. Wei L and Yuying Y. Multi-objective optimization of an auto panel drawing die face design by mesh morphing. *Comput Aided Des* 2007; 39: 863–869.
14. Wei L, Yuying Y, Zhongwen X, et al. Springback control of sheet metal forming based on the response-surface method and multi-objective genetic algorithm. *Mater Sci Eng A* 2009; 499: 325–328.
15. Wei D, Cui Z and Chen J. Optimization and tolerance prediction of sheet metal forming process using response surface model. *Comput Mater Sci* 2008; 42(2): 228–233.
16. Kurra S, Nasih HR, Regalla S, et al. Parametric study and multi-objective optimization in single-point incremental forming of extra deep drawing steel sheets. *Proc IMechE, Part B: J Engineering Manufacture* 2015; 230(5): 825–837.
17. Azamirad G and Arezoo B. Topology optimization of stamping die components using evolutionary structural optimization method. *Proc IMechE, Part B: J Engineering Manufacture* 2015; 231(4): 690–698.
18. Aminzadeh A, Parvizi A and Moradi M. Multi-objective topology optimization of deep drawing dissimilar tailor laser welded blanks; experimental and finite element investigation. *Opt Laser Technol* 2020; 125: 106029.
19. Modi B and Kumar DR. Optimization of process parameters to enhance formability of AA 5182 alloy in deep drawing of square cups by hydroforming. *J Mech Sci Technol* 2019; 33(11): 5337–5346.
20. Yixiong F, Lu R, Gao Y, et al. Multi-objective optimization of VBHF in sheet metal deep-drawing using Kriging, MOABC, and set pair analysis. *Int J Adv Manuf Technol* 2018; 96: 3127–3138.
21. Kitayama S, Yokoyama M, Kawamoto K, et al. Practical approach of simultaneous optimization of variable blank holder force and variable slide velocity trajectory in sheet metal forming. *Int J Adv Manuf Technol* 2018; 98: 2693–2703.
22. Qiu L, Zhang S, Wang Z, et al. A robust optimization design method for sheet metal roll forming and its application in roll forming circular cross-section pipe. *Int J Adv Manuf Technol* 2019; 103: 2903–2916.
23. Hughes TJR, Cottrell JA and Bazilevs Y. Isogeometric analysis: CAD, finite elements, NURBS, exact geometry and mesh refinement. *Comput Methods Appl Mech Eng* 2005; 194(39–41): 4135–4195.
24. Wall WA, Frenzel MA and Cyron C. Isogeometric structural shape optimization. *Comput Methods Appl Mech Eng* 2008; 197(33–40): 2976–2988.
25. Xu J, Gao L, Xiao M, et al. Isogeometric topology optimization for rational design of ultra-lightweight architected materials. *Int J Mech Sci* 2020; 166: 105103.
26. Liu Z, Wang C, Duan G, et al. A new refined plate theory with isogeometric approach for the static and buckling analysis of functionally graded plates. *Int J Mech Sci* 2019; 161–162: 105036.
27. Schulte J, Dittmann M, Eugster SR, et al. Isogeometric analysis of fiber reinforced composites using Kirchhoff–Love shell elements. *Comput Methods Appl Mech Eng* 2020; 362: 112845.
28. Benson DJ, Bazilevs Y, Hsu MC, et al. A large deformation, rotation-free, isogeometric shell. *Comput Methods Appl Mech Eng* 2011; 200(13–16): 1367–1378.
29. Zhang X, Zhu X, Wang C, et al. Initial solution estimation for one-step inverse isogeometric analysis in sheet metal stamping. *Comput Methods Appl Mech Eng* 2018; 330: 629–645.
30. Wang C, Zhang X, Shen G, et al. One-step inverse isogeometric analysis for the simulation of sheet metal forming. *Comput Methods Appl Mech Eng* 2019; 349: 458–476.
31. Shamloofard M and Assempour A. Development of an inverse isogeometric methodology and its application in sheet metal forming process. *Appl Math Model* 2019; 73: 266–284.
32. Shamloofard M and Assempour A. Simulation of sheet metal forming processes by presenting a bending-dependent inverse isogeometric methodology. *Int J Adv Manuf Technol* 2021; 112: 1389–1408.
33. Isazadeh AR, Shamloofard M and Assempour A. Some improvements on the one-step inverse isogeometric analysis by proposing a multi-step inverse isogeometric methodology in sheet metal stamping processes. *Appl Math Model* 2021; 91: 476–492.
34. Piegl L and Tiller W. *The NURBS book*. Berlin, Heidelberg: Springer Berlin Heidelberg, 1997.
35. Hashemi R, Mamusi H and Masoumi A. A simulation-based approach to the determination of forming limit diagrams. *Proc IMechE, Part B: J Engineering Manufacture* 2014; 228(12): 1582–1591.

36. Holmberg S and Thilderkvist P. Influence of material properties and stamping conditions on the stiffness and static dent resistance of automotive panels. *Mater Des* 2002; 23: 681–691.
37. Mitchell M. An Introduction to Genetic Algorithms. Cambridge, MA : The MIT Press, 1998.
38. Berger PD, Maurer RE and Celli GB. Introduction to Taguchi methods. In: *Experimental design*. Cham, Switzerland: Springer International Publishing, 2018, pp.449–480.
39. ASM Handbook Committee. ASM Metals Handbook. Vol. 14 Forming and Forging. Metals Park, Ohio: ASM International, 1988.